

CONNECTION DESIGN ADVISOR

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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Acknowledged. As STRUCTMIND AI, I will provide optimal connection recommendations for a 3-storey SMF + OCBF mixed system. The recommendations prioritize performance, economy, and constructability, referencing AISC 360-22, AISC 341-22 (Seismic Provisions), and AISC 358-22 (Prequalified Connections).

1. Beam-Column Joints (SMF System)

These connections are part of the Special Moment Frame and must be highly ductile, accommodating significant inelastic rotation. Their design is governed by AISC 341 and AISC 358.

- Recommended Connection Type: Bolted End Plate (BSEP), specifically the 4-bolt or 8-bolt unstiffened extended configuration per AISC 358, Chapter 6. This is a robust, prequalified connection with excellent performance.
- Governing Limit States (AISC 360-22 & 358-22):
 - Bolts:
 - Bolt Tensile Rupture (J3.6) - Governs bolts near tension flange.
 - Bolt Shear Rupture (J3.6) - Governs bolts near beam web.
 - End Plate:
 - Flexural Yielding of the End Plate (per AISC 358-22 eq. 6.8-1).
 - Welds (Shop Welds):
 - Strength of CJP groove weld at beam tension flange to end plate.
 - Strength of fillet welds at beam web to end plate (J2.4).
 - Column (Parent Member):
 - Column Panel Zone Shear Yielding (J10.6) - Often critical in SMF design.
 - Column Flange Local Bending (J10.1).
 - Column Web Local Yielding (J10.2).

- Beam (Connected Member):

- The design intent is for the beam to form a plastic hinge. The connection must be designed to be stronger than the beam's expected plastic moment, $M_{pe} = C_{pr} R_y F_y Z_x^*$.

- Bolt / Weld Sizing Guidance:

- Bolts: Use high-strength ASTM F3125 Gr. A325 or A490 bolts. Sizing is prescriptive per AISC 358 tables based on beam depth and flange thickness. Typically 1" to 1-1/2" diameter.

- Welds: Beam flanges are connected to the end plate with CJP groove welds. The beam web is connected with fillet welds designed to develop the full shear capacity of the web.

- Constructability Score: 8/10

- Remarks: Excellent choice for constructability. All welding is performed in the controlled shop environment. Field erection involves only bolting, which is fast, tolerant, and requires standard inspection. Reduces field CJP welding and associated NDT costs/delays.

- Alternate Solutions:

1. Reduced Beam Section (RBS): A classic prequalified connection where portions of the beam flanges are trimmed to create a "fuse" away from the column face. Requires more complex shop fabrication.

2. Welded Unreinforced Flange - Welded Web (WUF-W): The traditional field-welded moment connection. Constructability is lower (5/10) due to extensive field CJP welding, weather sensitivity, and rigorous UT inspection requirements.

2. Brace-to-Gusset Connection (OCBF System)

This connection must transfer high axial forces from the brace into the frame and accommodate brace buckling out-of-plane. Design is governed by AISC 341, Chapter F.

- Recommended Connection Type: Bolted Extended Gusset Plate. The gusset plate is shop-welded to the beam/column, and the brace (typically HSS) is field-bolted to the gusset.

- Governing Limit States (AISC 360-22 & 341-22):

The connection must be designed for the required strength, which is the lesser of the expected brace tensile strength ($R_y F_y A_g$) or the maximum force that can be developed by the system. Brace Member: Tensile Rupture on Net Section (J4.1b). Block Shear Rupture (J4.3). Bolts: Bolt Shear Rupture (J3.6). Bolt Bearing on Brace and Gusset Plate (J3.10) - Often governs due to thinner HSS walls or gusset plates. Gusset Plate: Tensile Yielding on Whitmore Section (J4.1a). Shear Yielding/Rupture on plate edges (J4.2). Block Shear Rupture (J4.3). Buckling of the Gusset Plate (J4.4) - The Uniform Force Method (UFM) and special geometric constraints from AISC 341 must be used. A clearance of at least $2t$ (where t^* is gusset thickness) must be provided for brace buckling.

- Bolt / Weld Sizing Guidance:

- Bolts: Size bolts to transfer the full required brace strength. Arrange bolt pattern to maximize block shear and bearing capacity.
- Welds: The shop welds connecting the gusset to the beam/column must also be designed to transfer the full required brace strength. Use balanced weld design to minimize eccentricity.
- Constructability Score: 9/10
- Remarks: This is the industry standard for braced frames. It separates complex shop fabrication (welding) from simple field erection (bolting), leading to high speed and reliability on site. Detailing must be precise to ensure fit-up.
- Alternate Solutions:
 1. Welded Brace-to-Gusset: The brace is directly field-welded to the gusset. This lowers constructability (6/10) due to field welding complexity but can result in a more compact connection.

3. Column Base Plates

The base plate type depends on the framing system it supports. SMF columns require fixed bases; OCBF columns require bases that can transfer high uplift/compression and shear.

- Recommended Connection Type:
 - For SMF Columns: Exposed Moment Base Plate. This base plate is designed as fixed, transferring moment, shear, and axial load into the foundation via anchor rods.
 - For OCBF Columns: High-Axial Load Pinned Base. While idealized as "pinned," it must resist significant tension (uplift) and compression from brace action, as well as frame shear.
- Governing Limit States (AISC 360-22, Ch. J & ACI 318):
 - Concrete & Anchors (per ACI 318):
 - Concrete Bearing Strength (AISC J8).
 - Anchor Rod Tensile Strength.
 - Concrete Breakout/Pullout/Side-Face Blowout Strength in Tension.
 - Anchor Rod Shear Strength.
 - Concrete Pryout/Edge Breakout Strength in Shear.
 - Base Plate:
 - Flexural Yielding of the plate under bearing pressure and anchor tension (AISC Design Guide 1).
 - Welds:
 - Strength of welds connecting column to base plate (J2).

- Bolt / Weld Sizing Guidance:

- Anchor Rods: Use ASTM F1554 Gr. 36, 55, or 105. Sizing is an iterative process based on the M-P-V interaction at the column base. Often, concrete breakout strength governs the design, requiring deeper embedment or anchor reinforcement.

- Welds: Use fillet or PJP groove welds to attach the column to the base plate, sized to transfer the full column forces.

- Constructability Score: 7/10

- Remarks: A standard but critical interface. Success depends heavily on the accurate placement of anchor rods by the concrete trade. Templates are essential. Grouting procedure is also a key quality control step. Minor misalignments can be costly to fix.

- Alternate Solutions:

1. Embedded Column Base: The column is embedded directly into the foundation. Offers excellent fixity but complicates foundation construction.

2. Proprietary High-Strength Base Connections: Systems that use special grout and anchoring hardware to achieve fixity with smaller footprints.

4. Column Splices

Splices allow columns to be fabricated and shipped in manageable lengths (typically two stories). They are usually located away from the beam-column joint.

- Recommended Connection Type: Bolted Flange Plate Splice. Located approx. 4 ft above the finished floor for ease of access during erection.

- Governing Limit States (AISC 360-22 & 341-22):

- AISC 341-22 Section D2.5 Requirements:

- Must be designed for the required shear and axial forces at the splice location.

- In SMF columns, the splice must also develop at least 50% of the flexural strength ($F_y Z$) of the smaller column section.

- Limit States:

- Bolt Shear Rupture (J3.6).

- Bolt Bearing on Flanges and Splice Plates (J3.10).

- Gross Section Yielding / Net Section Rupture of Splice Plates (J4.1).

- Block Shear Rupture of Splice Plates (J4.3).

- Bolt / Weld Sizing Guidance:

- Bolts: Use bearing-type connections. Size bolts and plates to meet the shear, axial, and 50% moment requirement from AISC 341.

- Shop Prep: Column ends must be milled for full contact bearing to transfer compression loads.

- Constructability Score: 9/10

- Remarks: Extremely common, efficient, and well-understood by fabricators and erectors. Erection tabs and shims are often used to aid alignment before final bolting.

- Alternate Solutions:

1. Complete Joint Penetration (CJP) Welded Splice: Provides full moment and axial capacity but is very expensive and difficult to execute in the field. Requires highly skilled labor, extensive inspection, and is weather-dependent. Low constructability (4/10).

5. Beam Splices (Gravity System)

Note: Splices are generally prohibited within the protected zones of SMF beams. These recommendations apply to gravity beams framing into the seismic system or other non-seismic members.

- Recommended Connection Type: Bolted Web Plate Shear Splice. If some moment continuity is required, it can be supplemented with bolted flange plates.

- Governing Limit States (AISC 360-22, Ch. J):

- Bolts:

- Bolt Shear Rupture (J3.6).

- Bolt Bearing on Beam Web and Splice Plates (J3.10).

- Splice Plates:

- Shear Yielding / Shear Rupture (J4.2).

- Flexural Yielding / Flexural Rupture (if transferring moment).

- Block Shear Rupture (J4.3).

- Bolt / Weld Sizing Guidance:

- Design the splice to transfer the shear and moment values at the splice location, typically placed at a point of low moment to optimize the connection.

- For a simple shear splice, ensure the bolt group can resist any inherent eccentricity.

- Constructability Score: 8/10

- Remarks: Standard and efficient method for extending beam lengths beyond stock sizes or shipping limits. Simple to erect and inspect.

- Alternate Solutions:

1. All-Welded Splice: Can be done in the shop to create a longer custom beam from shorter stock lengths, but rarely done in the field for gravity beams due to cost.